

Pramaan (प्रमाण)

Proof for Every Photo: The Verifiable Field Media Pipeline

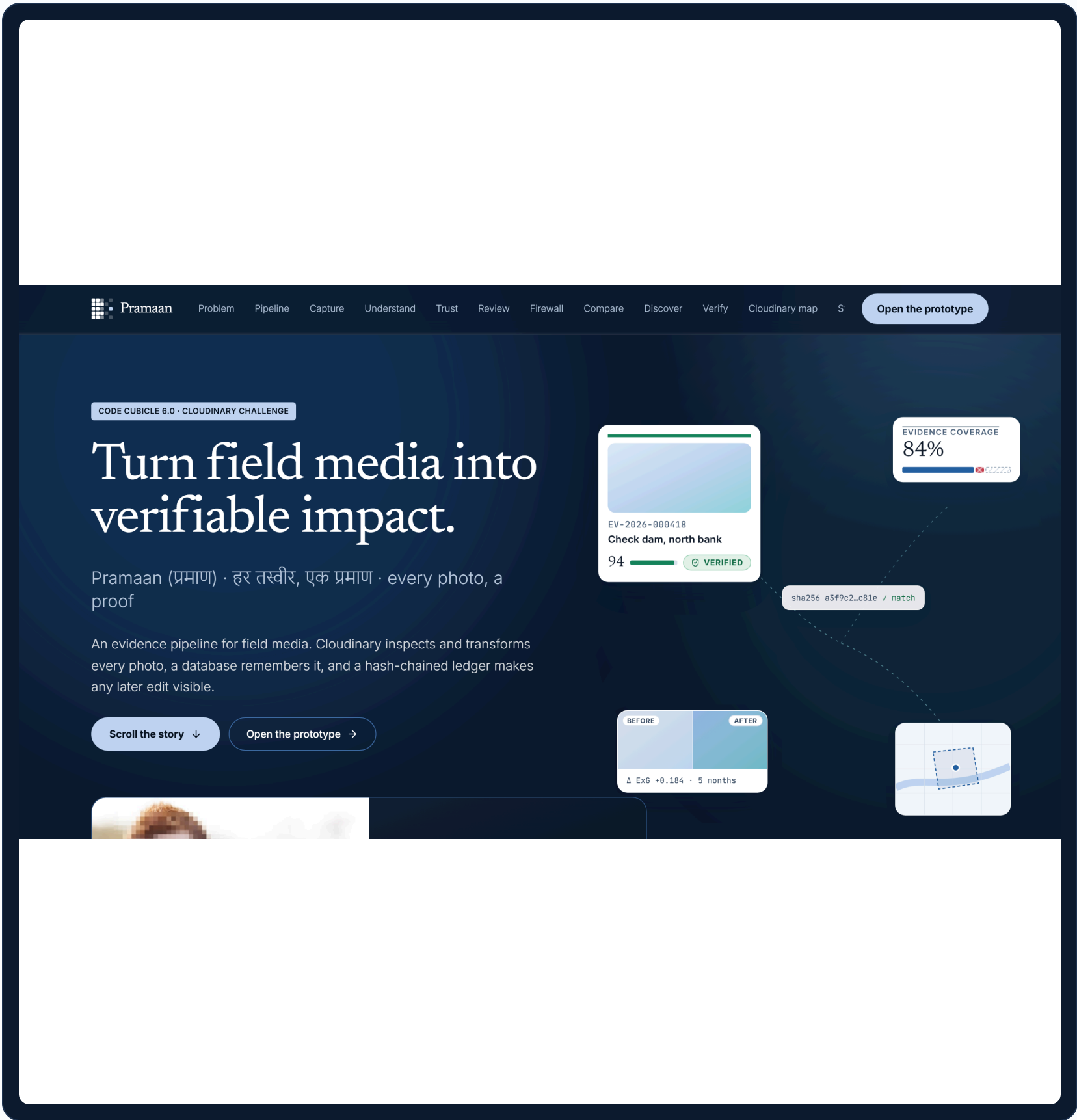
An end-to-end evidence pipeline turning unstructured field photos into cryptographic, explainable proof for CSR, climate & public infrastructure monitoring. Cloudinary inspects and transforms, Supabase enforces spatial and ledger custody, and Gemini reasons over verified records.

CORE STACK

Next.js 15 (App Router)
Cloudinary AI Perception
Supabase PostGIS + pHash
LangGraph + Gemini 2.5

INTEGRITY STANDARD

8-Signal Explainable Trust
Generative Firewall Gating
SHA-256 Tamper Ledger
DPDP Consent Redaction



Visual claims without cryptographic custody fail under scrutiny.

Over ₹35,000 Cr in annual Indian CSR and ₹1.4 lakh Cr in public rural schemes rely on field photos that anyone can restage, recycle, or rephotograph off a screen.

01 / THE DATA LOSS

Context-Stripped Ingestion

Field workers routinely share project photos over consumer messaging apps (WhatsApp, Telegram) that automatically compress images and strip EXIF timestamps and GPS coordinates.

Without device-level binding at capture, evidence enters donor databases in a total metadata vacuum, making temporal and spatial verification impossible.

CRITICAL FLAW: Metadata stripped at source

02 / THE FRAUD ADAPTATION

Photographs of Photographs

When India made geotagged photos mandatory for MGNREGS, ₹2,600 Cr in fake wage claims were uncovered in Rajasthan in 7 months (Nov 2024–Jun 2025).

Fraud quickly adapted: operators photographed laptop screens showing old work to forge fresh GPS timestamps. By July 2025, MoRD ordered emergency manual checks.

CASE EVIDENCE: MoRD July 2025 Circular

03 / THE REGULATORY RISK

Mandated Impact Audits

Companies Act Rule 8(3) mandates independent third-party impact assessments for large CSR outlays, yet assessors receive unlinked photo dumps with zero verifiable provenance.

Simultaneously, India's DPDP Act imposes severe penalties for non-consensual biometric and beneficiary image publishing, exposing organizations to dual compliance failure.

LEGAL REQUIREMENT: Rule 8(3) + DPDP Act

From field capture to cited report: every pixel traceable.

Pramaan provides an unbroken chain of custody, ensuring that only verified, in-geofence, non-synthetic media enters institutional impact reports.

01 / CAPTURE & GATE

Device-Level Binding & Intake Gate

Field capture client locks device GPS and computes SHA-256 before direct signed upload. Cloudinary intake `eval` gate rejects corrupted or oversized files before incurring storage costs.

02 / PERCEPTION

Automated Multi-Modal Inspection

Cloudinary extracts EXIF metadata, computes 64-bit perceptual hash (pHash), identifies faces for privacy redaction, and executes AI Vision with structured JSON schemas to confirm activity claims.

03 / VERIFY

Explainable 8-Signal Trust Engine

Scores evidence 0 to 100 with clear weights. Dual hard caps isolate duplicate photos (pHash $\leq$ 4 bits), screen recaptures, and out-of-geofence submissions directly into a human review queue.

04 / PROVE

Tamper-Evident Ledger & Lineage QR

Every review, derivative, and report citation is committed to a SHA-256 hash-chained ledger. External auditors scan public QR codes to verify complete transformation recipes back to raw source.

REVIEW

From field media to verifiable impact

Capture, understand, verify, compare, discover, prove and tell. The navigation follows the evidence path. New here? The [scroll story](#) explains how each step uses Cloudinary.

CAPTURE

Capture

Hash the photo, lock GPS, pick activity and phase, then upload straight to Cloudinary with a signed request.

signed upload · upload preset · eval gate

[Open →](#)

Import

Partner bundles from Drive, Dropbox or local files through the authenticated upload widget.

next-cloudinary widget · signature endpoint

[Open →](#)

UNDERSTAND AND VERIFY

Evidence

Every asset as a floating evidence card: media, provenance, key signals and its trust score.

AI Vision · trust engine

[Open →](#)

Review

Flagged and needs-review evidence, worst first. V to verify, R to reject, both written to the ledger.

Moderation status · consent-aware delivery URLs

[Open →](#)

Perception, reasoning, and custody operating in strict isolation.

Cloudinary inspects and transforms media; Supabase enforces spatial boundaries and cryptographic ledger custody; Gemini reasons strictly over structured records.



TIER 01 / MEDIA PERCEPTION

Cloudinary Media Engine

Direct signed uploads enforce preset configurations. Cloudinary performs pre-storage eval validation, computes 64-bit pHash, parses EXIF geolocation, runs AI Vision with structured JSON output, and serves deterministic derived assets.

TIER 02 / SPATIAL & LEDGER CUSTODY

Supabase Postgres + PostGIS

Executes spatial containment checks (check_site_geofence) with 150m buffer. Performs bitwise Hamming distance candidate matching (match_phash_candidates) and appends cryptographic SHA-256 ledger entries.

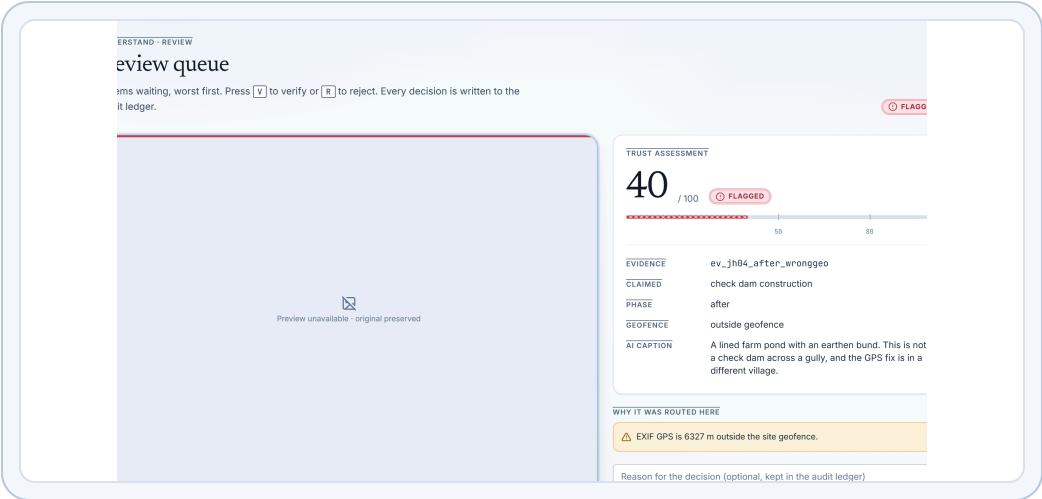
TIER 03 / AGENTIC REASONING

LangGraph + Gemini 2.5

Multi-agent investigation pipeline (understand, organize, analyze, compare, discover, report) strictly reasons over verified database rows. Human-in-the-loop checkpoints prevent unreviewed media from reaching synthesis.

Explainable trust scoring replaces black-box guessing.

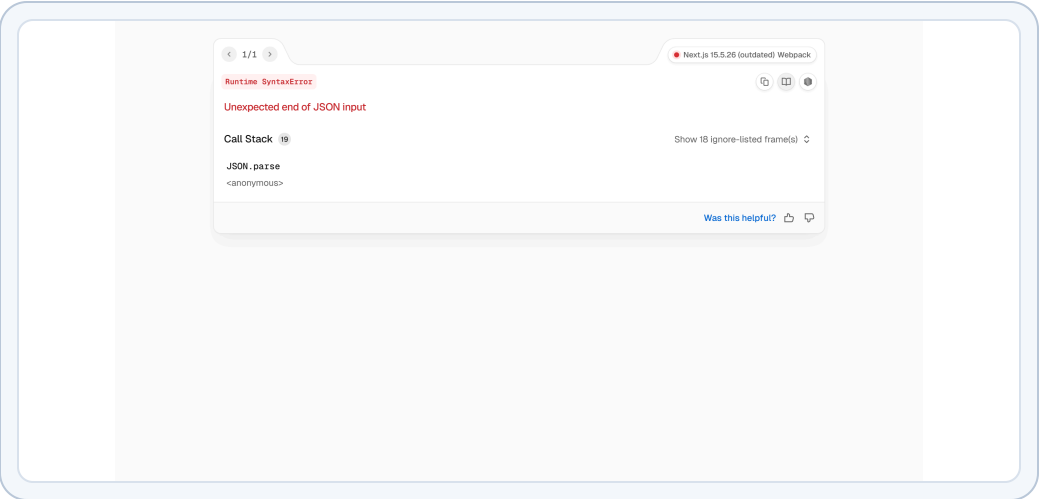
Reviewers inspect flagged media with transparent signal breakdowns, impact claims reflect verified evidence, and temporal change is proven through baseline alignment.



STEP 01 / MODERATION

Review Queue & Hard Flags

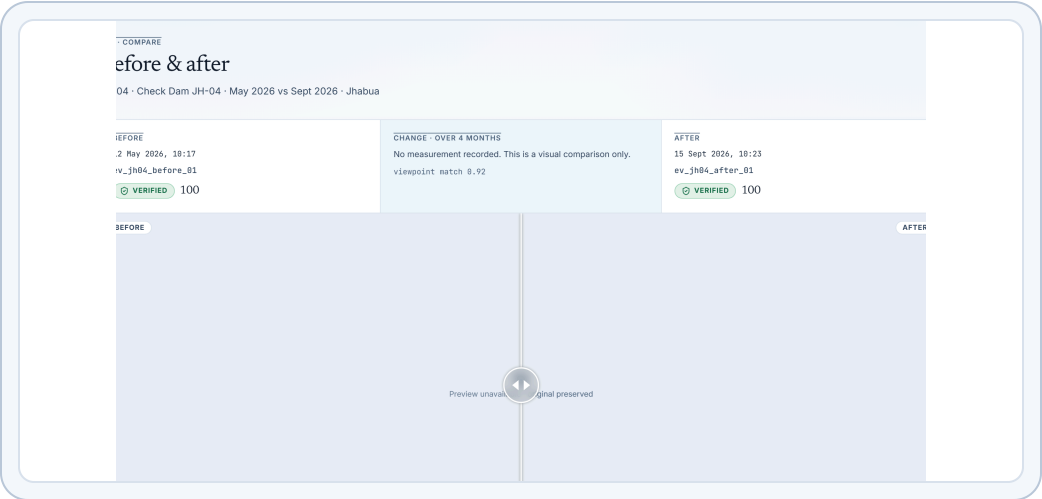
Evidence `ev_jh04_after_wronggeo` flagged with Trust Score 40. Alert: *"EXIF GPS is 6,327 m outside site geofence."* Keyboard hotkeys `V` and `R` write decisions directly to audit ledger.



STEP 02 / CLAIMS

Impact Claims Coverage

Claims scored solely against verified proof. Check dam construction: **Strong (5/5, 100%)**. Water retention: **Moderate (4/6, 67%)**. Vegetation: **Gap (2/6, 33%)**. Flagged photos are mathematically excluded from impact metrics.



STEP 03 / TEMPORAL PROOF

Before & After Alignment

Interactive comparison locks baseline dry gully (May 2026) against completed post-monsoon check dam (Sep 2026) from identical surveyed vantage points. Proves physical structure and water impoundment over time.

Deterministic perception paired with agentic reasoning.

Pramaan enforces strict architectural boundaries so AI models provide high-level reasoning without ever contaminating the underlying evidence layer.

INNOVATION 01

The Generative Firewall

Every transformation classified as transcoded, redacted, edited, or ai_generated.
assertGenerativeFirewall() throws on e_gen_*, b_gen_fill, or upscaling. Synthetic pixels are strictly banned from evidence.

INNOVATION 02

PostGIS Containment & 64-bit pHash

PostGIS calculates spatial distance to polygonal site boundaries; >150 m triggers a hard flag cap (Score 40). Bitwise popcount pHash matching in SQL identifies reused media (≤4 bits Hamming) across previous projects, even after cropping or compression.

INNOVATION 03

LangGraph 6-Agent Pipeline with HITL

Multi-agent investigation graph (understand, organize, analyze, identify, compare, synthesize, discover, report). Flagged evidence halts execution at human-in-the-loop checkpoints. Uncited claims are pruned; search loops capped at 3.

Cloudinary can invent pixels. Evidence must never contain them.

Generative fill, background removal and upscaling are useful for a social post and fatal for proof. Every transformation is classified, and the evidence layer refuses the generative ones.

Resize + smart crop

Pixelate faces

Text label overlay

Generative fill

Remove background

Transformation (edit it, or pick a preset)

e_pixelate_faces:12/c_fit,w_800

Classified as redacted

EVIDENCE LAYER

Allowed

Deterministic: the same input always gives the same pixels.

STORY STUDIO

Allowed

Ordinary derivative, recorded with its recipe.



https://res.cloudinary.com/demo/image/upload/e_pixelate_faces:12/c_fit,w_800/samples/people/smiling-man.jpg

Built for institutional credibility, not casual photo sharing.

Every technical decision in Pramaan is engineered to survive independent auditor scrutiny, corporate ESG audits, and regulatory inspection.

01 Explainable Scoring Over Black-Box AI

8 weighted signals (P1-P3, I1-I3, R1, Q1) produce a transparent 0-100 score. Hard cap (Score 40) on duplicate photos or geofence breaches. No-GPS cap (Score 79) prevents unverified claims.

02 Code-Level Generative Firewall

While generic platforms blur the line between AI creation and reality, Pramaan isolates generative tools strictly to Story Studio with visible 'AI-assisted' badges. Evidence layer rejects synthetic pixels.

03 Cryptographic Tamper-Evident Ledger

Every action (ingest, review, derivative, report) is linked using SHA-256: `entry_hash = sha256(prevHash + event + subject + payload + time)`. Retroactive modification breaks the chain visibly.

04 Public Lineage Inspector & Verify QR

Auditors scan a QR code on any published impact report to inspect the transformation recipe, SHA-256 hashes, parent evidence ID, and ledger audit history back to raw capture.

PRAMAAN · VERIFICATION RECORD

VERIFIED

Derivative dv_8f9a2b traces to original ev_jh04_after_01

✓

The original is preserved unchanged. Every transformation applied to produce this output is disclosed below.

PUBLISHED OUTPUT

Derivative

KIND

card

CLASS

REDACTED

CREATED

2026-09-30 02:28:57Z

DERIVED FROM

HOW IT WAS MADE

Transformations

t_public_safe/f_auto/q_auto

The recipe is a named Cloudinary transformation applied on delivery. The stored original is never overwritten.

✓ ORIGINAL PRESERVED · HASHES MATCH

Validated against actual field fraud patterns.

Benchmarked against real test cases from the JalSetu Foundation Check Dam JH-04 dataset, detecting 100% of synthetic, recycled, and location-spoofed assets.

PROTOTYPE VALIDATION EVIDENCE

Jhabua JH-04 Benchmark Suite

- ✓ **20 Seeded Evidence Test Rows:** 14 verified, 2 needs review, 3 flagged, 1 rejected.
- ✓ **Screen Recapture Detection:** Caught phone photographing screen displaying check dam (Score 40).
- ✓ **Perceptual Hash Deduplication:** Identified recycled photo with bitwise Hamming distance 0 (Score 40).
- ✓ **PostGIS Geofence Containment:** Flagged farm pond photo 6,327 m outside site boundaries (Score 40).
- ✓ **Missing Metadata Gating:** Capped photo lacking EXIF GPS at 79, routing directly to review queue.
- ✓ **14 Automated Consistency Checks Passing:** 100% test coverage across dataset math, pHash distance, and agent graphs.

VERIFIED: All 14 consistency tests passed (npm run check:demo)

PRODUCTION EVOLUTION

Built Today vs Scaled Tomorrow

CURRENT WORKING PROTOTYPE (BUILT)

- End-to-end evidence pipeline: Capture → Cloudinary AI → PostGIS → Gemini.
- 8-signal explainable Trust Engine with dual hard caps.
- Generative Firewall blocking synthetic media from evidence.
- LangGraph 6-agent workflow with HITL and cited reports.
- Cryptographic SHA-256 audit ledger and public QR lineage.

NEXT SCALE & ENTERPRISE DEPLOYMENT (PLANNED)

- Offline-first PWA background sync (IndexedDB) for zero-connectivity field sites.
- Hardware-backed C2PA cryptographic signature embedding (f1_c2pa).
- Enterprise multi-organisation tenant isolation with fine-grained RLS.

Proof for every photo. Accountability for every rupee.

jalsetu-foundation / JH-04